

Lecture Notes on Linear Discriminant Analysis (LDA)

1 Introduction to Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis (LDA) is a powerful technique used for both **dimensionality reduction** and **classification**. The primary objective of LDA is to find a linear transformation that maximizes the separation between different classes in a dataset. Unlike PCA, which focuses on preserving variance without considering class labels, LDA explicitly takes class information into account.

2 Goals of LDA

The main goals of LDA are:

- **Maximize class separability:** LDA finds a projection that maximizes the distance between different classes.
- **Reduce dimensionality:** LDA projects the data onto a lower-dimensional subspace, typically one dimension less than the number of classes.
- **Facilitate classification:** By maximizing the separation between classes, LDA enhances the discriminative power of features.

3 Mathematical Foundations of LDA

LDA involves finding a transformation that projects the data onto a new axis where the separation between classes is maximized. This is achieved by maximizing the **between-class scatter** and minimizing the **within-class scatter**.

3.1 Step 1: Calculating the Mean Vectors

For a dataset with C classes, let $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be the set of n data points, each belonging to one of C classes.

Define the mean vector of each class c as:

$$\mathbf{m}_c = \frac{1}{n_c} \sum_{\mathbf{x}_i \in \text{class } c} \mathbf{x}_i$$

where n_c is the number of samples in class c . The overall mean vector of the dataset is:

$$\mathbf{m} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$$

3.2 Step 2: Calculating the Scatter Matrices

To find a projection that maximizes class separability, LDA uses two scatter matrices:

- **Within-Class Scatter Matrix** (S_W): Measures the scatter (spread) within each class.
- **Between-Class Scatter Matrix** (S_B): Measures the scatter between the different classes.

Within-Class Scatter Matrix (S_W)

The within-class scatter matrix is defined as:

$$S_W = \sum_{c=1}^C \sum_{\mathbf{x}_i \in \text{class } c} (\mathbf{x}_i - \mathbf{m}_c)(\mathbf{x}_i - \mathbf{m}_c)^\top$$

This matrix captures how much the samples in each class vary around their class mean.

Between-Class Scatter Matrix (S_B)

The between-class scatter matrix is defined as:

$$S_B = \sum_{c=1}^C n_c (\mathbf{m}_c - \mathbf{m})(\mathbf{m}_c - \mathbf{m})^\top$$

This matrix captures how much the class means vary around the overall mean.

3.3 Step 3: Formulating the Objective Function

LDA aims to find a transformation matrix W that maximizes the **Fisher criterion**, which is the ratio of the between-class scatter to the within-class scatter. Mathematically, the objective is:

$$W = \underset{W}{\operatorname{argmax}} \frac{|W^\top S_B W|}{|W^\top S_W W|}$$

where $|\cdot|$ denotes the determinant of a matrix.

3.4 Step 4: Solving the Generalized Eigenvalue Problem

To maximize the Fisher criterion, we can solve the following **generalized eigenvalue problem**:

$$S_W^{-1} S_B \mathbf{w} = \lambda \mathbf{w}$$

where λ represents the eigenvalues and $\mathbf{w}$ the corresponding eigenvectors.

The eigenvectors corresponding to the largest eigenvalues form the columns of the transformation matrix W . These eigenvectors define the new feature space that maximizes class separability.

4 Dimensionality of the LDA Subspace

The maximum number of dimensions in the LDA subspace is $C - 1$, where C is the number of classes. This is because the between-class scatter matrix S_B has rank at most $C - 1$. Therefore, LDA projects the data onto a $(C - 1)$ -dimensional subspace.

5 Intuition Behind LDA

The intuition behind LDA can be summarized as follows:

- **Maximizing Separation:** By maximizing the distance between class means (using S_B) and minimizing the spread within each class (using S_W), LDA enhances the separation between classes.
- **Projection onto a Subspace:** LDA finds a lower-dimensional space where the classes are most distinguishable.

6 Algorithmic Steps for LDA

The following are the main steps of the LDA algorithm:

1. Compute the mean vectors $\mathbf{m}_c$ for each class and the overall mean vector $\mathbf{m}$.
2. Compute the within-class scatter matrix S_W and the between-class scatter matrix S_B .
3. Solve the generalized eigenvalue problem $S_W^{-1}S_B\mathbf{w} = \lambda\mathbf{w}$.
4. Select the eigenvectors corresponding to the largest eigenvalues to form the transformation matrix W .
5. Transform the dataset $\mathbf{X}$ using W to obtain the lower-dimensional representation.

7 Example: Two-Class LDA

In the case of two classes, LDA reduces the data to a 1-dimensional space. The Fisher criterion can be simplified as:

$$J(\mathbf{w}) = \frac{(\mathbf{w}^\top(\mathbf{m}_1 - \mathbf{m}_2))^2}{\mathbf{w}^\top S_W \mathbf{w}}$$

The optimal direction $\mathbf{w}$ is the one that maximizes this criterion, which can be found by solving:

$$\mathbf{w} \propto S_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2)$$

8 Applications of LDA

LDA is commonly used in the following areas:

- **Classification:** LDA is frequently used as a dimensionality reduction technique before applying a classifier.
- **Face Recognition:** LDA is often used to reduce the dimensionality of face images, helping improve classification performance.
- **Speech Recognition:** LDA can be applied to speech data for reducing dimensions and enhancing discriminative features.

9 Conclusion

Linear Discriminant Analysis (LDA) is a valuable tool for both dimensionality reduction and classification. By maximizing between-class scatter and minimizing within-class scatter, LDA projects data onto a subspace that enhances class separability, making it useful for a wide range of applications in machine learning and data analysis.